The black mark (▬) indicates Starting Points.
SCT: Standard Completion Time (Min./Sheet)
This is the time in which the student should complete each worksheet, including time taken for corrections.
(SCT is not used in Levels 6A-5A and ZI-ZII.)

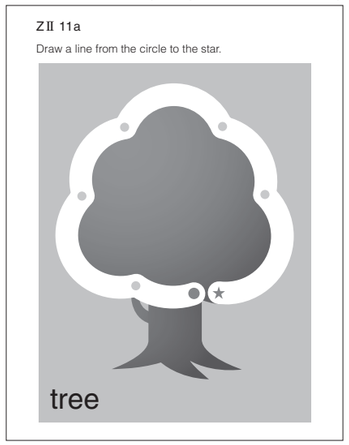
TABLE OF LEARNING MATERIALS I-1
MATHEMATICS (6A-2A) / PENCIL SKILLS PROGRAMME (ZI-ZII)

KUMON

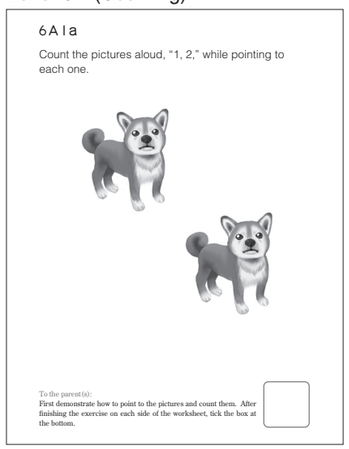
December 2012

	Pencil Skills Programme			6A	5A	4A	SCT	3A	SCT	2A	SCT
ZI	1- 10	Colouring 1	1- 10	Counting (Up to 5) 1	Number Reading Exercises (Up to 30) 1	Number Tracing Exercises 1	0.5–2	Numbers up to 100 Part 1	0.5–2	Review up to 3A	1–2
	11- 20	Colouring 2	11- 20	Counting (Up to 5) 2	Number Reading Exercises (Up to 30) 2	Number Tracing Exercises 2	0.5–2	Numbers up to 100 Part 2	0.5–2	Adding 4 Part 1 (Up to 12+4)	1–2
	21- 30	Straight Lines 1	21- 30	Counting (Up to 5) 3	Number Reading Exercises (Up to 30) 3	Number Tracing Exercises 3	0.5–2	Numbers up to 100 Part 3	0.5–2	Adding 4 Part 2 (Up to 16+4)	1–2
	31- 40	Straight Lines 2	31- 40	Counting (Up to 10) 1	Number Reading Exercises (Up to 30) 4	Number Tracing Exercises 4	0.5–2	Numbers up to 100 Part 4	0.5–2	Adding 5 Part 1 (Up to 12+5)	1–2
	41- 50	Straight Lines 3	41- 50	Counting (Up to 10) 2	Number Reading Exercises (Up to 30) 5	Number Writing Exercises up to 10 Part 1	0.5–2	Numbers up to 100 Part 5	0.5–2	Adding 5 Part 2 (Up to 15+5)	1–2
	51- 60	Curved Lines 1	51- 60	Counting (Up to 10) 3	Number Reading Exercises (Up to 30) 6	Number Writing Exercises up to 10 Part 2	0.5–2	Numbers up to 100 Part 6	0.5–2	Adding up to 5 Part 1	1–2
	61- 70	Curved Lines 2	61- 70	Counting (Up to 10) 4	Number Reading Exercises (Up to 30) 7	Number Writing Exercises up to 10 Part 3	0.5–2	Numbers up to 120	1–2	Adding up to 5 Part 2	1–2
	71- 80	Curved Lines 3	71- 80	Counting (Up to 10) 5	Number Reading Exercises (Up to 30) 8	Number Writing Exercises up to 10 Part 4	0.5–2	Adding 1 Part 1 (Up to 12+1)	1–2	Adding 6 Part 1 (Up to 12+6)	1–2
	81- 90	Curved Lines 4	81- 90	Counting (Up to 10) 6	Number Reading Exercises (Up to 30) 9	Number Writing Exercises up to 10 Part 5	0.5–2	Adding 1 Part 2 (Up to 18+1)	1–2	Adding 6 Part 2 (Up to 14+6)	1–2
	91-100	Curved Lines 5	91-100	Counting (Up to 10) 7	Number Reading Exercises (Up to 30) 10	Number Writing Exercises up to 10 Part 6	0.5–2	Adding 1 Part 3 (Up to 24+1)	1–2	Adding 7 Part 1 (Up to 11+7)	1–2
ZII	1- 10	Shapes and Pictures 1	101-110	Number Reading Exercises (Up to 10) 1	Sequence of Numbers (Up to 30) 1	Number Writing Exercises up to 20 Part 1	0.5–2	Adding 1 Part 4 (Up to 30+1)	1–2	Adding 7 Part 2 (Up to 13+7)	1–2
	11- 20	Shapes and Pictures 2	111-120	Number Reading Exercises (Up to 10) 2	Sequence of Numbers (Up to 30) 2	Number Writing Exercises up to 20 Part 2	0.5–2	Adding 1 Part 5 (Up to 60+1)	1–2	Adding up to 7 Part 1	1–2
	21- 30	Shapes and Pictures 3	121-130	Number Reading Exercises (Up to 10) 3	Sequence of Numbers (Up to 30) 3	Number Writing Exercises up to 30 Part 1	0.5–2	Adding 1 Part 6 (Up to 1000+1)	1–2	Adding up to 7 Part 2	1–2
	31- 40	Shapes and Pictures (Stories) 1	131-140	Number Reading Exercises (Up to 10) 4	Sequence of Numbers (Up to 40) 1	Number Writing Exercises up to 30 Part 2	0.5–2	Adding 2 Part 1 (Up to 14+2)	1–2	Adding 8 Part 1 (Up to 11+8)	1–2
	41- 50	Shapes and Pictures (Stories) 2	141-150	Number Reading Exercises (Up to 10) 5	Sequence of Numbers (Up to 40) 2	Numbers up to 50 Part 1	0.5–2	Adding 2 Part 2 (Up to 18+2)	1–2	Adding 8 Part 2 (Up to 12+8)	1–2
	51- 60	Shapes and Pictures (Stories) 3	151-160	Number of Dots (Up to 10) 1	Sequence of Numbers (Up to 40) 3	Numbers up to 50 Part 2	0.5–2	Adding 2 Part 3 (Up to 32+2)	1–2	Adding 9 (Up to 12+9)	1–2
	61- 70	Back and Forth 1	161-170	Number of Dots (Up to 10) 2	Sequence of Numbers (Up to 50) 1	Numbers up to 50 Part 3	0.5–2	Adding 3 Part 1 (Up to 14+3)	1–2	Adding 9 and 10 (Up to 12+9 and 15+10)	1–2
	71- 80	Back and Forth 2	171-180	Number of Dots (Up to 10) 3	Sequence of Numbers (Up to 50) 2	Numbers up to 50 Part 4	0.5–2	Adding 3 Part 2 (Up to 21+3)	1–2	Adding up to 10 Part 1	1–2
	81- 90	Corners and Curves 1	181-190	Number of Dots (Up to 10) 4	Sequence of Numbers (Up to 50) 3	Numbers up to 50 Part 5	0.5–2	Adding up to 3 Part 1	1–2	Adding up to 10 Part 2	1–2
	91-100	Corners and Curves 2	191-200	Number of Dots (Up to 10) 5	Large Numbers	Numbers up to 50 Part 6	0.5–2	Adding up to 3 Part 2	1–2	Adding up to 10 Part 3	1–2

Levels ZI & ZII (Shapes and Pictures)

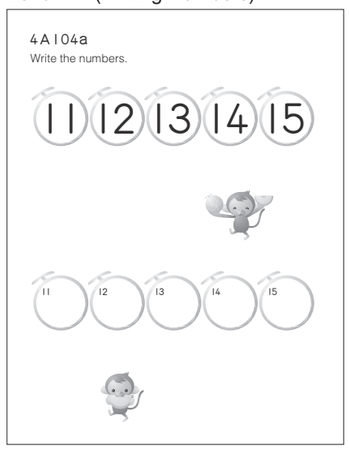


Level 6A (Counting)

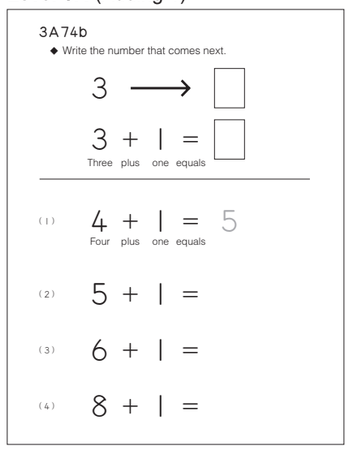


Level 5A (Reading Numbers)

Level 4A (Writing Numbers)



Level 3A (Adding 1)



Level 2A (Adding up to 10)

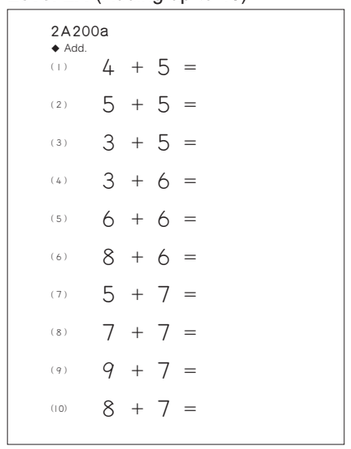


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TABLE OF LEARNING MATERIALS I-2
MATHEMATICS (A-F)

KUMON

October 2016

	A	SCT	B	SCT	C	SCT	D	SCT	E	SCT	F	SCT
1- 10	Addition 1 (Review up to 2A)	1–2	Addition (Review up to A)	1–2	Review up to B	2–3	Review up to C	2–3	Review up to D	2–3	Review up to E 1	3–5
11- 20	Addition 2 (Up to sum of 12)	1–2	Addition to 100 Part 1	1–2	Multiplication up to 3	2–3	Multiplication: 2 Digits × 2 Digits 1	2–4	Addition of Fractions 1	2–3	Review up to E 2	3–5
21- 30	Addition 3 (Up to sum of 15)	1–2	Addition to 100 Part 2	2–3	Multiplication up to 5	2–3	Multiplication: 2 Digits × 2 Digits 2	2–4	Addition of Fractions 2	3–4	Multiplication and Division of 3 Fractions	4–6
31- 40	Addition 4 (Up to sum of 18)	1–2	Addition to 100 Part 3	2–3	Multiplication up to 7	2–3	Multiplication: 2 Digits × 2 Digits 3	3–4	Addition of Fractions 3	3–4	Addition of 3 Fractions 1	3–5
41- 50	Addition 5 (Up to sum of 20)	1–2	Addition of 2-Digit Numbers 1	2–3	Multiplication up to 9	2–3	Multiplication: 3 Digits × 2 Digits	3–5	Addition of Fractions 4	3–4	Addition of 3 Fractions 2	4–6
51- 60	Addition 6 (Up to sum of 24)	1–2	Addition of 2-Digit Numbers 2	2–3	Multiplication: 2 Digits × 1 Digit 1	2–4	Addition and Subtraction	2–4	Addition of Fractions 5	3–5	Addition and Subtraction of 3 Fractions	4–6
61- 70	Addition 7 (Up to sum of 28)	1–2	Addition of 2-Digit Numbers 3	2–3	Multiplication: 2 Digits × 1 Digit 2	2–4	Multiplication and Division 1	3–4	Addition of Fractions 6	3–5	Order of Operations (3 Fractions) 1	3–5
71- 80	Addition 8 (Summary of addition)	2–3	Addition of 3-Digit Numbers 1	2–4	Multiplication: 2 Digits × 1 Digit 3	2–4	Multiplication and Division 2	3–4	Addition of Fractions 7	3–5	Order of Operations (3 Fractions) 2	3–5
81- 90	Subtraction 1 (Subtracting 1)	1–2	Addition of 3-Digit Numbers 2	2–4	Multiplication: 2 Digits × 1 Digit 4	2–4	Division by 2-Digit Numbers 1	3–4	Addition of Fractions 8	3–5	Order of Operations (3 Fractions) 3	3–5
91-100	Subtraction 2 (Subtracting 2)	1–2	Addition of 3-Digit Numbers 3	3–5	Multiplication: 2 Digits × 1 Digit 5	2–4	Division by 2-Digit Numbers 2	3–5	Addition of Fractions 9	4–6	Order of Operations (3-or-More Fractions) 1	4–6
101-110	Subtraction 3 (Subtracting 3)	1–2	Subtraction 1 (Review up to A)	1–2	Multiplication: 3 or 4 Digits × 1 Digit	3–5	Division by 2-Digit Numbers 3	3–5	Subtraction of Fractions 1	3–4	Order of Operations (3-or-More Fractions) 2	4–6
111-120	Subtraction 4 (Subtracting up to 3)	1–2	Subtraction 2 (Review up to A)	2–3	Introduction to Division	2–3	Division by 2-Digit Numbers 4	3–5	Subtraction of Fractions 2	3–5	Order of Operations (3-or-More Fractions) 3	4–6
121-130	Subtraction 5 (Subtracting up to 5)	1–2	Subtraction of 2-Digit Numbers 1	2–3	Division with Remainders 1	2–3	Division by 2-Digit Numbers 5	3–5	Subtraction of Fractions 3	3–5	Order of Operations (3-or-More Fractions) 4	4–6
131-140	Subtraction 6 (From numbers up to 10)	1–2	Subtraction of 2-Digit Numbers 2	2–3	Division with Remainders 2	2–3	Division: Quotients of 2-or-More Digits 1	3–5	Addition and Subtraction of Fractions	4–6	Fractions and Decimals 1	3–5
141-150	Subtraction 7 (From numbers up to 11)	1–2	Subtraction of 2-Digit Numbers 3	2–3	Division with Remainders 3	2–3	Division: Quotients of 2-or-More Digits 2	4–6	Multiplication of Fractions 1	3–4	Fractions and Decimals 2	4–6
151-160	Subtraction 8 (From numbers up to 12)	1–2	Addition and Subtraction of 2-Digit Numbers	2–4	Division with Remainders 4	2–3	Fractions	3–5	Multiplication of Fractions 2	3–5	Fractions and Decimals 3	4–6
161-170	Subtraction 9 (From numbers up to 14)	1–2	Subtraction of 3-Digit Numbers 1	2–3	Division: 2 Digits ÷ 1 Digit 1	2–3	Reduction 1	2–3	Division of Fractions	3–5	Solving Word Problems 1	4–6
171-180	Subtraction 10 (From numbers up to 16)	1–2	Subtraction of 3-Digit Numbers 2	2–4	Division: 2 Digits ÷ 1 Digit 2	2–3	Reduction 2	2–3	Multiplication and Division of Fractions	3–5	Solving Word Problems 2	5–7
181-190	Subtraction 11 (From numbers up to 20)	1–2	Subtraction of 3-Digit Numbers 3	2–4	Division: 3 Digits ÷ 1 Digit 1	3–4	Reduction 3	2–4	Four Operations of 2 Fractions 1	3–5	Decimals 1	3–5
191-200	Subtraction 12 (Summary of subtraction)	2–3	Subtraction of 3-Digit Numbers 4	3–5	Division: 3 Digits ÷ 1 Digit 2	3–5	Reduction 4	2–4	Four Operations of 2 Fractions 2	3–5	Decimals 2	3–5

Level A (Subtraction)

A200a
◆ Subtract.
(1) 10 − 3 =
(2) 11 − 6 =
(3) 12 − 7 =
(4) 15 − 4 =
(5) 16 − 8 =
(6) 13 − 7 =
(7) 17 − 9 =
(8) 14 − 5 =
(9) 18 − 6 =
(10) 19 − 2 =
(11) 20 − 3 =

Level B (Addition to 100)

B21a
◆ Add.
(1) 15 + 5 = □□
(2) 25 + 5
(3) 35 + 5
(4) 45 + 5
(5) 55 + 5
(6)* 15 + 15 = □□
(7) 25 + 15
(8) 35 + 15
(9) 45 + 15
(10) 55 + 15

Level C (Multiplication)

C52a
◆ Multiply.
(1) 32 × 2
(2) 42 × 2
(3) 52 × 2
(4) 62 × 2
(5) 72 × 2
(6) 53 × 3
(7) 63 × 3
(8) 73 × 3
(9) 83 × 3
(10) 93 × 3

Level D (Division)

D81a
◆ Divide.
(1) □ R 3; 21) 45; □ ← 21×2; 3 ← 45 − 42
(2) □ R □; 21) 47; □
(3) □ R □; 21) 48; □□; □
(4) 21) 49
(5) □ R □; 21) 65; □□; □
(6) 21) 67
(7) 21) 68
(8) 21) 69

Level E (Addition of Fractions)

E70a
◆ Add.
(1) $\frac{3}{8} + \frac{2}{9} =$
(2) $\frac{3}{4} + \frac{3}{14} =$
(3) $\frac{3}{10} + \frac{9}{16} =$
(4) $\frac{7}{10} + \frac{11}{20} =$
(5) $\frac{5}{6} + \frac{4}{21} =$
(6) $\frac{7}{8} + \frac{7}{24} =$

Level F (Order of Operations)

F121a
◆ Calculate.
(1) $\left(\frac{2}{3} - \frac{1}{4}\right) \times 6$ =
(2) $\frac{3}{8} \times 4 - \frac{3}{10}$ =
(3) $\frac{5}{6} \div \frac{1}{3} - 1\frac{1}{10}$ =
(4) $1\frac{1}{5} \times \frac{2}{3} - 1\frac{2}{3} \div 2\frac{2}{9}$ =

TABLE OF LEARNING MATERIALS II-1 (Level G to L)

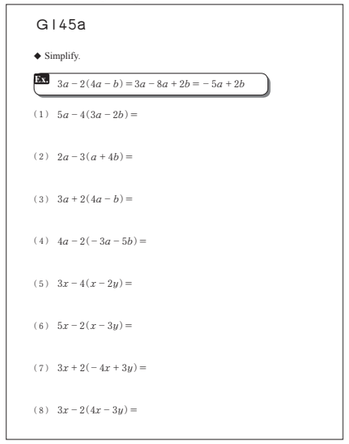
KUMON

The black mark (▬) indicates most suitable Starting Points.
SCT: Standard Completion Time (Min./Sheet)
This is the set time in which the student should complete each worksheet, including time taken for correction.

October 2016

	G	SCT	H	SCT	I	SCT	J	SCT	K	SCT	L	SCT
1- 10	▬ Review up to F 1	3–5	▬ Basics for Level H Mathematics 1	4–6	▬ Basics for Level I Mathematics	4–6	▬ Expansion of Polynomial Products	5–8	Review of Linear Functions	4–6	Logarithmic Functions	6–12
11- 20	Review up to F 2	3–5	Basics for Level H Mathematics 2	4–6	Multiplication of Polynomials	4–6	Factorisation I	5–8	Review of Quadratic Functions	5–8	Graphs of Logarithmic Functions	7–14
21- 30	Addition and Subtraction of Positive and Negative Numbers 1	2–4	Literal Equations 1	4–6	Multiplication Using Formulas	4–6	Factorisation II	5–8	Quadratic Functions and Graphs	6–12	Logarithmic Equations and Inequalities	8–16
31- 40	Addition and Subtraction of Positive and Negative Numbers 2	3–5	Literal Equations 2	4–6	▬ Factorisation 1	4–6	Factorisation III	6–10	Determining Equations of Quadratic Functions	7–14	Modulus Functions	8–16
41- 50	Addition and Subtraction of Positive and Negative Numbers 3	3–5	Simultaneous Equations in Two Variables 1	4–6	Factorisation 2	4–6	Factorisation IV	6–10	Maxima and Minima of Quadratic Functions I	7–14	Limits and Derivatives	8–16
51- 60	Addition and Subtraction of Positive and Negative Numbers 4	4–6	Simultaneous Equations in Two Variables 2	4–6	Factorisation 3	4–6	Factorisation V	7–12	Maxima and Minima of Quadratic Functions II	7–14	Tangents	12–24
61- 70	Multiplication of Positive and Negative Numbers	3–5	Simultaneous Equations in Two Variables 3	4–6	Factorisation 4	4–6	Fractional Expressions	6–10	Maxima and Minima of Quadratic Functions III	8–16	Relative Maxima and Minima I	15–30
71- 80	Division of Positive and Negative Numbers	4–6	Simultaneous Equations in Two Variables 4	4–6	Factorisation 5	4–6	Irrational Numbers I	5–8	Quadratic Functions and Equations	7–14	Relative Maxima and Minima II	15–30
81- 90	Four Operations with Positive and Negative Numbers 1	4–6	Simultaneous Equations in Two Variables 5	4–6	Square Roots 1	4–6	Irrational Numbers II	6–10	Quadratic Functions and Inequalities	7–14	Maxima and Minima I	15–30
91-100	Four Operations with Positive and Negative Numbers 2	4–6	Simultaneous Equations in Three and Four Variables 1	4–6	Square Roots 2	4–6	Quadratic Equations I	5–8	Quadratic Functions and Solutions of Quadratic Equations	8–16	Maxima and Minima II	15–30
101-110	▬ Values of Algebraic Expressions 1	4–6	Simultaneous Equations in Three and Four Variables 2	4–6	Square Roots 3	4–6	Quadratic Equations II	6–10	Higher Degree Functions	6–12	Applications to Equations and Inequalities	15–30
111-120	Values of Algebraic Expressions 2	4–6	Application of Equations	4–6	Quadratic Equations 1	4–6	Quadratic Equations and Complex Numbers	6–10	Higher Degree Equations and Inequalities	7–14	Indefinite and Definite Integrals	8–16
121-130	Simplifying Algebraic Expressions 1	3–5	Inequalities 1	4–6	Quadratic Equations 2	4–6	Discriminant	6–10	Graphs of Fractional Functions I	7–14	Definite Integrals I	12–24
131-140	Simplifying Algebraic Expressions 2	3–5	Inequalities 2	4–6	Quadratic Equations 3	4–6	Root-Coefficient Relationships	6–10	Graphs of Fractional Functions II	7–14	Definite Integrals II	12–24
141-150	Simplifying Algebraic Expressions 3	4–6	Functions and Graphs 1	4–6	Graphs of Quadratic Functions 1	4–6	Simultaneous Equations	6–10	Fractional Equations and Inequalities	8–16	Areas I	15–30
151-160	Simplifying Algebraic Expressions 4	4–6	Functions and Graphs 2	4–6	Graphs of Quadratic Functions 2	4–6	Dividing Polynomials	6–10	Graphs of Irrational Functions	7–14	Areas II	15–30
161-170	Linear Equations 1	3–5	Functions and Graphs 3	4–6	Graphs of Quadratic Functions 3	4–6	Remainder Theorem	6–10	Irrational Equations and Inequalities	8–16	Volumes	15–30
171-180	Linear Equations 2	4–6	Functions and Graphs 4	4–6	The Pythagorean Theorem 1	4–6	Factor Theorem	6–10	Exponential Functions	6–12	Velocity and Distance	15–30
181-190	Linear Equations 3	4–6	Simplifying Monomials and Polynomials 1	4–6	The Pythagorean Theorem 2	4–6	Proof of Identities	6–10	Graphs of Exponential Functions	7–14	Summary of Differentiation and Integration I	30–60
191-200	Linear Equations 4	4–6	Simplifying Monomials and Polynomials 2	4–6	The Pythagorean Theorem 3	4–6	Proof of Inequalities	7–12	Exponential Equations and Inequalities	8–16	Summary of Differentiation and Integration II	30–60

Level G
(Simplifying Algebraic Expressions)


G145a

◆ Simplify.

Ex. $3a-2(4a-b)=3a-8a+2b=-5a+2b$

(1) $5a-4(3a-2b)=$

(2) $2a-3(a+4b)=$

(3) $3a+2(4a-b)=$

(4) $4a-2(-3a-5b)=$

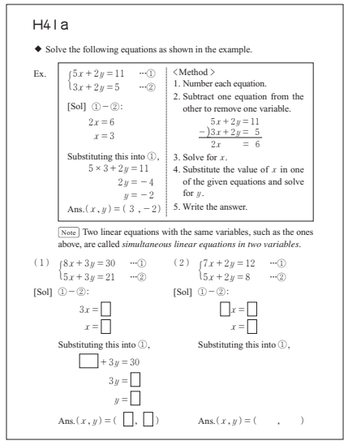
(5) $3x-4(x-2y)=$

(6) $5x-2(x-3y)=$

(7) $3x+2(-4x+3y)=$

(8) $3x-2(4x-3y)=$

Level H
(Simultaneous Equations)


H41a

◆ Solve the following equations as shown in the example.

Ex. $\begin{cases} 5x+2y=11 & \cdots① \\ 3x+2y=5 & \cdots② \end{cases}$

[Sol] ①−②:
$2x=6$
$x=3$
Substituting this into ①,
$5\times3+2y=11$
$2y=-4$
$y=-2$
Ans. $(x, y)=(3, -2)$

<Method>
1. Number each equation.
2. Subtract one equation from the other to remove one variable.
$$\begin{array}{r} 5x+2y=11 \\ -)\ 3x+2y=5 \\ \hline 2x\quad\ =6 \end{array}$$
3. Solve for x.
4. Substitute the value of x in one of the given equations and solve for y.
5. Write the answer.

Note Two linear equations with the same variables, such as the ones above, are called *simultaneous linear equations in two variables*.

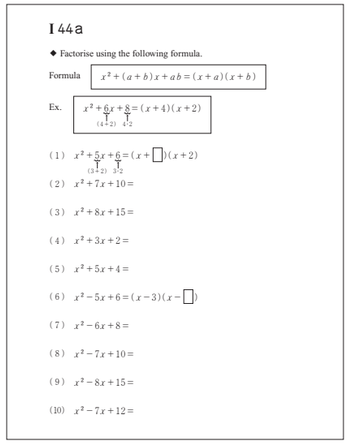
(1) $\begin{cases} 8x+3y=30 & \cdots① \\ 5x+3y=21 & \cdots② \end{cases}$

[Sol] ①−②:
$3x=\square$
$x=\square$
Substituting this into ①,
$\square+3y=30$
$3y=\square$
$y=\square$
Ans. $(x, y)=(\square, \square)$

(2) $\begin{cases} 7x+2y=12 & \cdots① \\ 5x+2y=8 & \cdots② \end{cases}$

[Sol] ①−②:
$\square x=\square$
$x=\square$
Substituting this into ①,
Ans. $(x, y)=(\quad, \quad)$

Level I
(Factorisation)


I44a

◆ Factorise using the following formula.

Formula $x^2+(a+b)x+ab=(x+a)(x+b)$

Ex. $x^2+\underset{(4+2)}{6}x+\underset{4\cdot2}{8}=(x+4)(x+2)$

(1) $x^2+\underset{(3+2)}{5}x+\underset{3\cdot2}{6}=(x+\square)(x+2)$

(2) $x^2+7x+10=$

(3) $x^2+8x+15=$

(4) $x^2+3x+2=$

(5) $x^2+5x+4=$

(6) $x^2-5x+6=(x-3)(x-\square)$

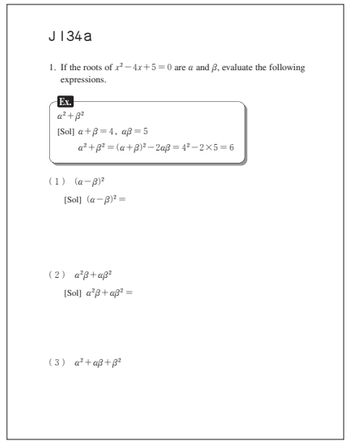
(7) $x^2-6x+8=$

(8) $x^2-7x+10=$

(9) $x^2-8x+15=$

(10) $x^2-7x+12=$

Level J
(Root-Coefficient Relationships)


J134a

1. If the roots of $x^2-4x+5=0$ are α and β, evaluate the following expressions.

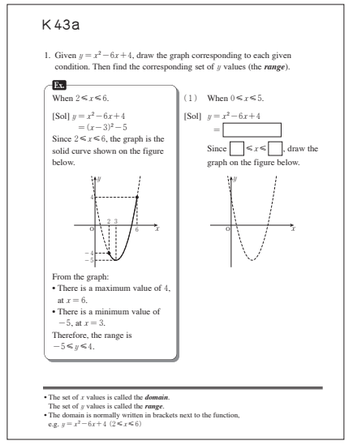
Ex. $\alpha^2+\beta^2$
[Sol] $\alpha+\beta=4$, $\alpha\beta=5$
$\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=4^2-2\times5=6$

(1) $(\alpha-\beta)^2$
[Sol] $(\alpha-\beta)^2=$

(2) $\alpha^2\beta+\alpha\beta^2$
[Sol] $\alpha^2\beta+\alpha\beta^2=$

(3) $\alpha^2+\alpha\beta+\beta^2$

Level K
(Maxima and Minima of Quadratic Functions)


K43a

1. Given $y=x^2-6x+4$, draw the graph corresponding to each given condition. Then find the corresponding set of y values (the ***range***).

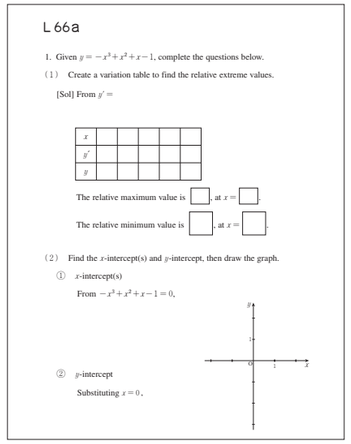
Ex. When $2\leqq x\leqq6$.
[Sol] $y=x^2-6x+4$
$=(x-3)^2-5$
Since $2\leqq x\leqq6$, the graph is the solid curve shown on the figure below.

From the graph:
• There is a maximum value of 4, at $x=6$.
• There is a minimum value of -5, at $x=3$.
Therefore, the range is $-5\leqq y\leqq4$.

(1) When $0\leqq x\leqq5$.
[Sol] $y=x^2-6x+4$
$=\square$
Since $\square\leqq x\leqq\square$, draw the graph on the figure below.

• The set of x values is called the ***domain***.
The set of y values is called the ***range***.
• The domain is normally written in brackets next to the function, e.g. $y=x^2-6x+4$ $(2\leqq x\leqq6)$

Level L
(Relative Maxima and Minima)


L66a

1. Given $y=-x^3+x^2+x-1$, complete the questions below.

(1) Create a variation table to find the relative extreme values.
[Sol] From $y'=$

x					
y'					
y					

The relative maximum value is $\square$, at $x=\square$.
The relative minimum value is $\square$, at $x=\square$.

(2) Find the x-intercept(s) and y-intercept, then draw the graph.
① x-intercept(s)
From $-x^3+x^2+x-1=0$,

② y-intercept
Substituting $x=0$,

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TABLE OF LEARNING MATERIALS II-2 (Level M to X)

KUMON

SCT: Standard Completion Time (Min./Sheet)
This is the set time in which the student should complete each worksheet, including time taken for correction.

March 2018

	M	SCT	N	SCT	O	SCT
1- 10	Points and Lines I	10–20	Arithmetic Sequences	10–20	Tangents and Normals	20–40
11- 20	Points and Lines II	10–20	Geometric Sequences	10–20	Increasing/Decreasing Functions and Relative Extreme Values	25–50
21- 30	Points and Lines III	15–30	Various Sequences I	15–30	Concavity of Curves	25–50
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41- 50	Circles II	15–30	Recurrence Relations	20–40	Various Applications of Differentiation	25–50
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61- 70	Loci II	15–30	Infinite Sequences	10–20	Indefinite Integrals II	15–30
71- 80	Regions	15–30	Infinite Geometric Sequences	15–30	Integration by Substitution	15–30
81- 90	Trigonometric Ratios I	10–20	Infinite Geometric Series	15–30	Integration by Parts	20–40
91-100	Trigonometric Ratios II	10–20	Infinite Series	15–30	Definite Integrals	15–30
101-110	Properties of Trigonometric Functions I	10–20	Limits of Functions I	15–30	Integration by Substitution for Definite Integrals	15–30
111-120	Properties of Trigonometric Functions II	10–20	Limits of Functions II	15–30	Integration by Parts for Definite Integrals and Functions Represented by Definite Integrals	20–40
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161-170	Addition Formulas II	15–30	Differentiation of Trigonometric Functions	15–30	Differential Equations	25–50
171-180	Addition Formulas III	15–30	Differentiation of Logarithmic and Exponential Functions	15–30	Natural/Social Science and Differential Equations	30–60
181-190	Sine and Cosine Rules	15–30	Differentiation of Various Functions and Higher Order Derivatives	20–40	Applications of Calculus I	30–60
191-200	Area of Triangles	15–30	Various Properties of Derivatives	25–50	Applications of Calculus II	30–60

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	XS	SCT
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Level M
(Cosine Rule)

M184a

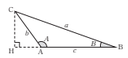
As shown in the diagram below, when A and B are both acute angles, prove that $a^2=b^2+c^2-2bc\cos A$.

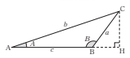
[Sol] In △ABC, drop a perpendicular CH from vertex C to AB. In △ACH,
CH = [ ], AH = [ ]
Also, BH=AB−AH = [ ]
In △BCH,
a^2=CH²+BH²
=([ ])²+([ ])²
$=b^2+c^2-2bc\cos A$
Therefore, $a^2=b^2+c^2-2bc\cos A$ is true.

$\sin^2 A+\cos^2 A=1$

When A is an obtuse angle,
CH $=b\sin(180^\circ-A)=b\sin A$
BH = AB + AH
$=c+b\cos(180^\circ-A)$
$=c-b\cos A$
a^2=CH²+BH²
$=(b\sin A)^2+(c-b\cos A)^2$
$=b^2+c^2-2bc\cos A$

When B is an obtuse angle,
CH $=b\sin A$
BH = AH − AB
$=b\cos A-c$
a^2=CH²+BH²
$=(b\sin A)^2+(b\cos A-c)^2$
$=b^2+c^2-2bc\cos A$





Therefore, when either A or B is an obtuse angle, $a^2=b^2+c^2-2bc\cos A$.
It is also true when either A or B is a right angle.
Similarly, $b^2=c^2+a^2-2ca\cos B$ and $c^2=a^2+b^2-2ab\cos C$ are true.
This is called the ***Cosine Rule***.

Level N
(Differentiation of Logarithmic Functions)

N171a

Using the definition of the derivative, differentiate $y=\log_a x$. $(x>0)$

[Sol] $y'=\lim_{h\to 0}\frac{\log_a(x+h)-\log_a x}{h}$ ← $\log_a M-\log_a N=\log_a\frac{M}{N}$

$=\lim_{h\to 0}\frac{1}{h}\log_a\frac{x+h}{x}$

$=\lim_{h\to 0}\frac{1}{h}\log_a\left(1+\frac{h}{x}\right)$

Let $\frac{h}{x}=k$. As $h\to 0$, $k\to 0$; therefore,

$y'=\lim_{k\to 0}\frac{1}{kx}\log_a(1+k)$ ← $n\log_a M=\log_a M^n$

$=\lim_{k\to 0}\frac{1}{\square}\log_a(1+k)^{\frac{1}{k}}$

As $k\to 0$, $(1+k)^{\frac{1}{k}}$ is known to approach a constant value, which is expressed as e.

So, $e=\lim_{k\to 0}(1+k)^{\frac{1}{k}}$

Using this constant value e,

$y'=\frac{1}{\square}\log_a e=\frac{\log_a e}{\square\log_a a}=\frac{1}{\log_e a}$ …① ← $\log_a M=\frac{\log_e M}{\log_e a}$, $\log_e a=1$

When $a=e$, $\log_e a=\log_e e=1$; therefore, from ①, $y'=\frac{1}{x}$

When examining the value of $(1+k)^{\frac{1}{k}}$ by substituting k with a value which is close to 0, it approaches a constant value as shown on the right. e is an irrational number and $e=2.7182818\ldots$.

k	$(1+k)^{\frac{1}{k}}$	k	$(1+k)^{\frac{1}{k}}$
0.1	2.59374…	−0.1	2.86797…
0.01	2.70481…	−0.01	2.73199…
0.001	2.71692…	−0.001	2.71964…
0.0001	2.71814…	−0.0001	2.71841…
0.00001	2.71826…	−0.00001	2.71829…

Level O
(Integration by Parts)

O81a

The Product Rule
$\{f(x)g(x)\}'=f'(x)g(x)+f(x)g'(x)$ ← N144
can be rearranged into
$f(x)g'(x)=\{f(x)g(x)\}'-f'(x)g(x)$
If both sides are integrated, then the following formula is obtained.

Integration by Parts

$$\int f(x)g'(x)\,dx=f(x)g(x)-\int f'(x)g(x)\,dx$$

This method is called ***Integration by Parts***.

Find the following indefinite integrals using Integration by Parts.

Ex.

$\int x\cos x\,dx=\int x\,(\sin x)'\,dx$ ← $\int f(x)\,g'(x)\,dx$

$=x\sin x-\int(x)'\sin x\,dx$

$=x\sin x-\int\sin x\,dx$ ← $f(x)\,g'(x)-\int f'(x)\,g(x)\,dx$

$=x\sin x+\cos x+C$

(1) $\int x\sin x\,dx=$

Level XV
(Vectors and Figures)

XV99a

1. Given △ABC where $A=90^\circ$, let D and E be the midpoints of BC and CA respectively. Show that if AD ⊥ BE, AC $=\sqrt{2}$ AB.

[Sol] Let $\overrightarrow{AB}=\vec{a}$ and $\overrightarrow{AC}=\vec{b}$.
$\overrightarrow{BE}=$
$\overrightarrow{AD}=$
Since $\overrightarrow{BE}\cdot\overrightarrow{AD}=$ [ ].

2. Given △ABC, let M be the midpoint of side BC. Show that $\overline{AB}^2+\overline{AC}^2=2(\overline{AM}^2+\overline{BM}^2)$ using the following two methods.

[Sol 1] Let $\overrightarrow{AB}=\vec{a}$ and $\overrightarrow{AC}=\vec{b}$.
LHS =
RHS =

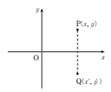
[Sol 2] Let $\overrightarrow{AM}=\vec{a}$ and $\overrightarrow{BM}=\vec{b}$.

Level XM
(Transformations)

XM62a

1. Given that P(x, y) is mapped to Q(x', y') by the reflection in the x-axis, find the matrix equation relating the two points.

[Sol] Since (x, y) is mapped to (x', y') by the reflection in the x-axis,
$x'=x$, $y'=$ [ ]
Rewriting,
$x'=1\cdot x+0\cdot y$
$y'=0\cdot x+$ [ ]
The matrix equation is:
$\begin{pmatrix}x'\\y'\end{pmatrix}=\begin{pmatrix}1 & 0\\ \square & \square\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$



2. Given that P(x, y) is mapped to Q(x', y') by the reflection in the origin, find the matrix equation relating the two points.

Note: In general, when S is a set, the mapping from S to S is called a transformation of S.

Level XP
(Permutations and Combinations)

XP24a

1. Find the number of ways 5 people A, B, C, D and E can sit around a round table.

[Sol] Examining 5 of the possibilities,

Since the seating relationship of the people in each of the above possibilities is the same,
all 5 possibilities show the same seating arrangement.
Therefore, if we were to use 5! to find the number of ways, it would be 5 times too large.
Thus, the total number of ways is: $\frac{5!}{\square}=$ [ ]

Number of Circular Permutations

The ways of arranging different objects in a ring are called ***circular permutations***.
The number of circular permutations of n objects is:
$\frac{n!}{n}=(n-1)!$

2. Find the number of ways 6 people can sit around a round table.

3. Find the number of ways 8 people can sit around a round table.